Team Alpha

Bop-It! Project Update Presentation

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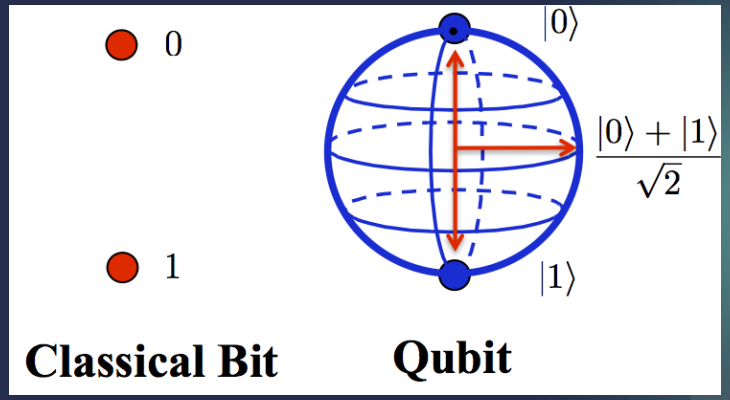
Our version of Bop-It….Qub-It!

Brief review of how a qubit works:

A qubit is like a spinning arrow (or coin):

- - Unlike a normal bit, it can be 0, 1, or both at once (superposition)
- Quantum gates rotate that arrow to control the qubit's state in 3D space in the x,y,z axes.
- When you measure, it collapses to either 0 or 1.
- Over time, noise can interfere, causing decoherence



Status Update

The DF-Player draws **too much current!**

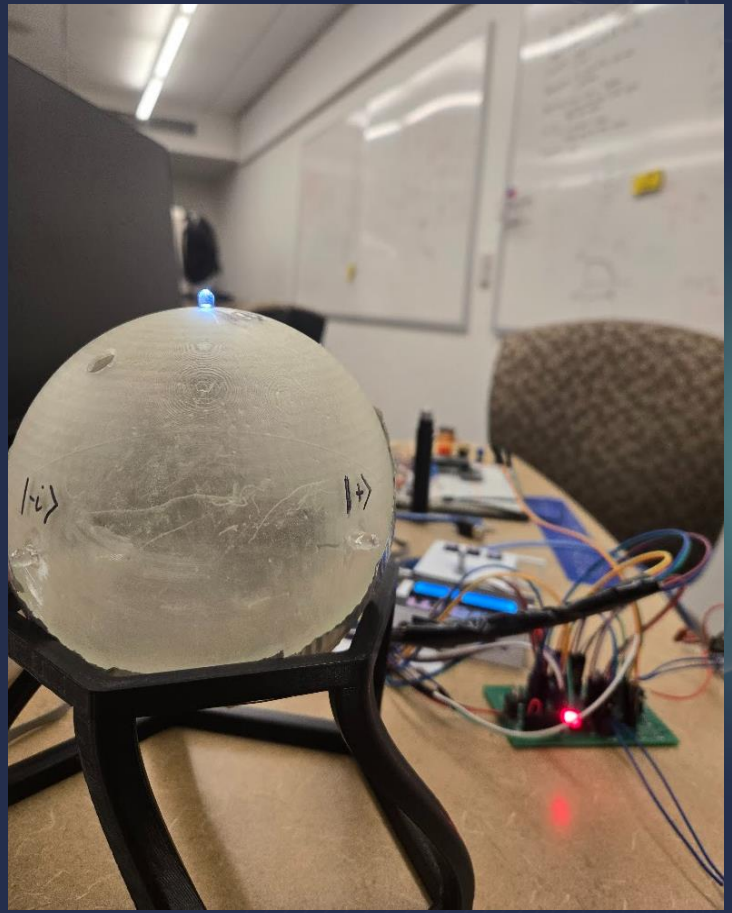
Our (2nd) PCB does not have the proper power distribution to handle high current draw.

A DF boot failure causes the rest of our code to not function properly

Status Update

Our enclosure demonstrates gating and measurement state transition functionality

However, our complete QUB-IT protocol only functions at the breadboard phase!



Future Work

- Third PCB redesign with appropriate power distribution
- Further enclosure iterations
  - More ergonomic
  - Fits new PCB
  - Snap-fit
  - Resin print lid
- New sphere with better diameter tolerance